Towards Generative World Modeling: A Comprehensive Survey for Novel View Synthesis

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Abstract

Novel View Synthesis (NVS)—the task of rendering a scene from arbitrary unseen viewpoints given a set of reference observations—has undergone a paradigm shift from classical geometry-based reconstruction toward learning-driven generative approaches. This survey examines NVS not merely as a rendering technique, but as the **foundational spatial capability** of Generative World Models: systems that predict visual scene states under arbitrary conditions, including viewpoint changes, temporal dynamics, and agent interactions. We provide a comprehensive review spanning three dominant paradigms—*implicit neural representations* (NeRF and variants), *explicit Gaussian-based representations* (3DGS and variants), and *generative model-driven synthesis* (diffusion models and score distillation methods)—and analyse how each advances, or falls short of, the full demands of world modelling. We further survey emerging 4D approaches that unify spatial and temporal prediction, and identify key open challenges including physical consistency, scalable generalisation, and the geometry–generation trade-off. Our analysis reveals that the field is at a critical inflection point where NVS and world modelling are converging, and we lay out a research roadmap toward truly generalisable, physically grounded world models.

1 Introduction

1.1 The Fundamental Challenge

Intelligence, whether biological or artificial, requires the ability to reason about the world beyond what is immediately visible. A surgeon planning an incision must mentally rotate a 3-D anatomy from a 2-D scan. A self-driving vehicle must anticipate occlusion from a new perspective before rounding a corner. A robot must predict how a grasped object will appear from the wrist camera before it reaches the target pose. In each case the core cognitive operation is the same: **synthesising a plausible visual state of the world at an unobserved vantage point**.

This is the problem of **Novel View Synthesis (NVS)**. Formally, given a set of reference images $\{\mathbf{I}_1, \ldots, \mathbf{I}_n\}$ captured from known camera poses $\{\boldsymbol{\pi}_1, \ldots, \boldsymbol{\pi}_n\}$, the goal is to render a photo-realistic image $\hat{\mathbf{I}}$ at an arbitrary target pose $\boldsymbol{\pi}^*$, faithfully preserving scene geometry, appearance, and lighting.

While the problem was articulated decades ago in the image-based rendering literature [1, 2], the past five years have witnessed an extraordinary leap in capability driven by deep learning, differentiable rendering, and generative modelling. Yet despite remarkable progress, a critical question remains underexplored: *in what sense does a system that can synthesise novel views "understand" the world?* And conversely, *what must NVS methods ultimately become in order to serve as the spatial backbone of a true World Model?*

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Table 1: Four eras of Novel View Synthesis characterised by dominant scene representation, input regime, and chief world-modelling limitation.

Era	Period	Representation	Input	Chief Limitation
I	2015–2020	Explicit mesh / light field	Dense multi-view	No learning prior; brittle
II	2020–2023	Implicit MLP (NeRF)	Dense $\rightarrow$ sparse	Per-scene; slow
III	2023–now	Explicit Gaussians (3DGS)	Dense; feed-fwd	Reconstruction-centric
IV	2022–now	Diffusion / score distillation	Single image/text	Geometry–generation gap

This survey is organised around these questions. We treat NVS not as an isolated rendering task, but as the **spatial prediction axis** of a larger problem: **Generative World Modelling**.

1.2 What Is a World Model?

> **Working Definition.** A *World Model* is a learned system that can predict the visual (and optionally physical) state of a scene under arbitrary conditions—including viewpoint changes, temporal evolution, and agent-driven interactions—in a manner that is *geometrically consistent*, *physically plausible*, and *generalisable* beyond the training distribution.

This definition deliberately extends the classic control-theoretic formulation [3, 4], where a world model predicts the next latent state given an action. We expand the "condition" axis to include camera pose as a first-class citizen, making NVS a natural special case: when the condition is a viewpoint change and the scene is static, the world model reduces exactly to an NVS system.

Conversely, a world model must go beyond NVS in two critical dimensions. (1) **Temporal prediction**: a world model must handle dynamic scenes, predicting how the scene evolves over time under physics, agent actions, or autonomous dynamics—static NVS methods cannot address this axis. (2) **Action conditioning**: a world model must be *controllable*; given an agent action (move forward, grasp, rotate), it must render the resulting observation, requiring semantic and physical understanding beyond geometric interpolation.

1.3 Why NVS Is the Core of World Modelling

Among all world model capabilities, **spatial consistency** is the foundational constraint. A temporal prediction that is geometrically inconsistent—where a coffee mug occupies two different positions when viewed from two angles—is not a world model; it is a hallucination. NVS is precisely the discipline that enforces multi-view geometric consistency, making it a non-negotiable prerequisite for any credible world model.

Central Thesis. *NVS is the spatial inference engine of world modelling. Without it, a "world model" reduces to a video generator—capable of producing plausible-looking sequences but unable to support downstream tasks that require spatial grounding, such as robot manipulation, surgical planning, or autonomous navigation.*

Advances in NVS—from dense input to sparse-view generalisation, from static scenes to dynamic 4-D modelling—map directly onto advances in world modelling capability. We use this lens throughout the survey to evaluate each method: not only by its PSNR on standard benchmarks, but by its contribution toward five world-modelling desiderata (Section 2).

1.4 The Evolution Arc: From Reconstruction to Generation

The history of NVS can be understood as a progressive relaxation of input constraints, each step demanding stronger scene priors and thus moving closer to generative world modelling (see Table 1).

Era I (2015–2020): Classical Reconstruction. Structure-from-Motion (SfM), multi-view stereo (MVS), and light-field methods dominated [7]. High-quality outputs were achievable under tightly controlled conditions, but no generative prior was available and dense input was mandatory.

Table 2: Capability coverage across NVS paradigms against world-model requirements. **+** achieved; $\sim$ partially/method-dependent; **–** not addressed.

Capability	**Classical**	**NeRF**	**3DGS**	**Gen. NVS**	**Target**
Geometric consistency	+	+	+	$\sim$	+
Sparse/single-image input	–	$\sim$	$\sim$	+	+
Dynamic scene (4-D)	–	$\sim$	$\sim$	$\sim$	+
Action/agent control	–	–	–	$\sim$	+
Physical plausibility	–	–	–	$\sim$	+
Cross-scene generalisation	–	$\sim$	$\sim$	+	+
Real-time rendering	$\sim$	–	+	–	+

Era II (2020–2023): Neural Radiance Fields. Mildenhall *et al.* [5] introduced NeRF: a continuous implicit scene representation encoded in an MLP, trained via differentiable volume rendering. The community rapidly extended NeRF to dynamic deformations [8, 10], unbounded scenes, and cross-scene generalisation [12, 13]. Quality surpassed classical methods by large margins; the key limitation was per-scene optimisation taking hours to days.

Era III (2023–present): 3D Gaussian Splatting. Kerbl *et al.* [19] replaced the implicit MLP with anisotropic Gaussian primitives rendered via differentiable rasterisation, enabling real-time, high-fidelity rendering. The community extended 3DGS to dynamic scenes [20, 21] and feed-forward generalisation [22, 23]. The paradigm remains reconstruction-centric with a weak generative prior.

Era IV (2022–present): Generative Model-Driven NVS. Diffusion models and score distillation methods [34, 29, 32, 33] enable NVS from single images or text prompts. Video diffusion models [39, 40, 38] unify temporal and viewpoint prediction, arriving at nascent 4-D World Models.

Each era coexists with the others, addressing different points on the trade-off frontier among reconstruction fidelity, input efficiency, generalisation, and computational cost. The current frontier lies in **integrating the geometric precision of NeRF/3DGS with the generative priors of diffusion models**, while extending both along the temporal axis—the path towards Generative World Modelling.

1.5 Existing Gaps

Table 2 summarises the capability landscape. No single existing paradigm satisfies all world-model requirements: reconstruction-based methods excel at geometric consistency but lack generative priors; generative methods provide strong priors but sacrifice geometric rigour.

1.6 Contributions

(1) **A World-Modelling Lens for NVS.** We introduce a unified analytical framework evaluating NVS methods on five world-modelling desiderata: geometric consistency, generalisation, dynamic modelling, physical plausibility, and action controllability.

(2) **Comprehensive Taxonomy Across Three Paradigms.** We survey more than 150 methods spanning classical reconstruction, NeRF variants, 3DGS variants, and generative NVS approaches, analysing each method's representation, input regime, rendering mechanism, and world-modelling implications.

(3) **The 4-D Convergence Thesis.** We articulate and substantiate the claim that the field is converging toward 4-D World Modelling—the unification of spatial (NVS) and temporal (video prediction) generation—and trace the technical developments driving this convergence.

(4) **Open Problems and Research Roadmap.** We identify the geometry–generation trade-off, long-horizon temporal consistency, physical grounding, and evaluation methodology as critical open problems, and propose a concrete research roadmap.

1.7 Paper Structure

Section 2 provides background: camera models, differentiable volume rendering, and the unified NVS formulation. Section 3 reviews implicit neural representations (NeRF). Section 4 covers explicit Gaussian representations (3DGS). Section 5 surveys generative NVS approaches. Section 6 synthesises these into the world-modelling perspective. Section 7 reviews datasets and benchmarks. Section 8 identifies open problems and presents the roadmap. Section 9 concludes.

2 Background and Problem Formulation

2.1 Camera Model

We adopt the standard pinhole camera model. A 3-D point $\mathbf{X} \in \mathbb{R}^3$ is projected to a 2-D pixel $\mathbf{p} \in \mathbb{R}^2$ via

$$\mathbf{p} = \mathbf{K}\,[\,\mathbf{R}\ \mathbf{t}\,]\,\tilde{\mathbf{X}}, \tag{1}$$

where $\mathbf{K}$ is the 3×3 intrinsic matrix, $\mathbf{R} \in \mathrm{SO}(3)$ and $\mathbf{t} \in \mathbb{R}^3$ define the extrinsic pose, and $\tilde{\mathbf{X}}$ denotes homogeneous coordinates. A camera ray emitted from the optical centre $\mathbf{o}$ along direction $\mathbf{d}$ is parameterised as $\mathbf{r}(s) = \mathbf{o} + s\,\mathbf{d}$, $s \in [s_n, s_f]$.

2.2 Volume Rendering

NeRF [5] represents a scene as a continuous radiance field $F_\theta : (\mathbf{x}, \mathbf{d}) \mapsto (\mathbf{c}, \sigma)$, where $\mathbf{c} \in \mathbb{R}^3$ is emitted colour and $\sigma \geq 0$ is volume density. The expected colour of a ray $\mathbf{r}$ is computed via the *volume rendering integral*:

$$\hat{C}(\mathbf{r}) = \int_{s_n}^{s_f} T(s)\,\sigma(\mathbf{r}(s))\ \mathbf{c}(\mathbf{r}(s), \mathbf{d})\ \mathrm{d}s, \tag{2}$$

where the transmittance $T(s) = \exp\!\Big(-\int_{s_n}^{s} \sigma(\mathbf{r}(u))\,\mathrm{d}u\Big)$ accounts for accumulated occlusion. In practice Eq. (2) is approximated by stratified sampling and quadrature.

2.3 Differentiable Rasterisation (3DGS)

3DGS [19] represents a scene as a set of N anisotropic 3-D Gaussians $\{\mu_i, \Sigma_i, \alpha_i, \mathbf{c}_i\}_{i=1}^{N}$, where $\mu_i \in \mathbb{R}^3$ is the mean, Σ_i the covariance, $\alpha_i \in [0, 1]$ the opacity, and $\mathbf{c}_i$ the (view-dependent) colour encoded as spherical harmonics. For a given viewpoint, each Gaussian is projected to a 2-D splat and composited front-to-back:

$$\hat{C} = \sum_{i \in \mathcal{N}} \mathbf{c}_i\,\alpha_i\,\tilde{G}_i \prod_{j<i} \Big(1 - \alpha_j\,\tilde{G}_j\Big), \tag{3}$$

where $\tilde{G}_i$ is the 2-D Gaussian evaluated at the pixel centre. Because all operations are differentiable, Σ_i, α_i, and $\mathbf{c}_i$ are optimised by gradient descent on photometric loss.

2.4 Unified NVS Formulation

We define NVS as the problem of learning a mapping

$$\mathcal{M}_\theta :\ \big(\{\mathbf{I}_i, \boldsymbol{\pi}_i\}_{i=1}^{n},\ \boldsymbol{\pi}^*\big)\ \longrightarrow\ \hat{\mathbf{I}}^*, \tag{4}$$

where n ranges from dense multi-view ($n \gg 1$) down to a single image ($n = 1$), and $\boldsymbol{\pi}^*$ is the target camera pose. The world-modelling extension augments Eq. (4) with a time index t and an action sequence $\mathbf{a}_{0:t}$:

$$\mathcal{M}_\theta :\ \big(\{\mathbf{I}_i, \boldsymbol{\pi}_i\}_{i=1}^{n},\ \mathbf{a}_{0:t},\ t,\ \boldsymbol{\pi}^*\big)\ \longrightarrow\ \hat{\mathbf{I}}_t^*. \tag{5}$$

This unified formulation makes explicit the five world-modelling desiderata: **(D1)** geometric consistency across $\boldsymbol{\pi}^*$; **(D2)** generalisation to unseen scenes ($n \leq 1$); **(D3)** dynamic scene modelling ($t > 0$); **(D4)** action conditioning ($\mathbf{a}_{0:t} \neq \varnothing$); **(D5)** physical plausibility.

3 Implicit Neural Representations: NeRF and Variants

3.1 Original NeRF

Mildenhall *et al.* [5] encode a static scene in the weights of an MLP F_θ that maps a 5-D input—3-D position $\mathbf{x}$ and 2-D viewing direction $\mathbf{d}$—to colour $\mathbf{c}$ and density σ. Positional encoding $\gamma(\mathbf{x})$ lifts inputs to a higher-frequency space, enabling the MLP to represent fine geometric detail. Training minimises a photometric reconstruction loss over held-out pixels via the volume rendering integral (Eq. 2).

World-Modelling Implications. NeRF achieves strong geometric consistency (D1) but satisfies none of the other desiderata: per-scene optimisation precludes generalisation (D2), the static assumption rules out dynamics (D3) and action conditioning (D4), and no physical prior is enforced (D5).

3.2 Acceleration Methods

Vanilla NeRF requires thousands of network evaluations per ray and hours of training. Three complementary strategies have been pursued.

Explicit grid hybrids. Plenoxels [15] and TensoRF [16] replace the MLP with sparse voxel grids or tensor decompositions, reducing training to minutes while maintaining quality.

Hash-based encodings. Instant-NGP [17] uses a multi-resolution hash grid to achieve sub-second training, enabling interactive scene capture.

Knowledge distillation / baking. Methods such as MobileNeRF [18] distil a trained NeRF into a mesh-texture representation, enabling real-time rendering on commodity hardware.

3.3 Generalisation Methods

The per-scene limitation of vanilla NeRF has been addressed by conditioning the network on image features extracted from reference views.

pixelNeRF [12] conditions F_θ on pixel-aligned image features from a CNN encoder, enabling single-image (D2) reconstruction. **IBRNet** [13] aggregates features from multiple reference views via cross-attention, generalising to novel scenes at test time. **MVSNeRF** [14] constructs a cost volume from multi-view stereo and conditions the radiance field on this volume, combining geometric priors with learned appearance.

3.4 Dynamic NeRF

Extending NeRF to dynamic scenes addresses desideratum D3.

D-NeRF [10] introduces a deformation network that maps each time-stamped point to a canonical space, where a standard NeRF is evaluated. **Nerfies** [8] and **HyperNeRF** [9] model more complex non-rigid deformations via latent deformation codes and higher-dimensional ambient spaces, respectively. **K-Planes** [11] decomposes the 4-D space-time radiance field into six orthogonal feature planes, enabling efficient optimisation of dynamic scenes.

World-Modelling Implications. Dynamic NeRF methods achieve D3, but per-scene optimisation still precludes generalisation (D2), and action conditioning (D4) remains absent.

4 Explicit Gaussian Representations: 3DGS and Variants

4.1 3D Gaussian Splatting

Kerbl *et al.* [19] initialise Gaussians from an SfM point cloud and optimise them jointly via the differentiable splatting render (Eq. 3). An adaptive densification strategy adds Gaussians in under-reconstructed regions and prunes low-opacity ones, yielding compact representations. Real-time rendering at $\sim$100 fps at 1080p is achieved on a modern GPU.

World-Modelling Implications. 3DGS achieves strong geometric consistency (D1) and real-time rendering, a prerequisite for interactive world models. However, like NeRF, it is per-scene optimised and lacks generative priors, leaving D2–D5 unaddressed.

4.2 Dynamic Extensions

4D Gaussian Splatting (4DGS) [20] extends Gaussians into a 4-D space-time representation, coupling each Gaussian with a time-varying deformation field. **SpacetimeGaussians** [21] parameterise Gaussian trajectories as polynomial functions of time, enabling smooth interpolation across frames. Both methods deliver real-time dynamic rendering, advancing D3 while retaining D1.

4.3 Feed-Forward Generalisation

Combining the efficiency of Gaussian splatting with the generalisation of learned priors has been a major recent thrust.

pixelSplat [22] predicts per-pixel Gaussian parameters from an image-pair encoder, enabling single-forward-pass 3DGS reconstruction for novel scene pairs (D2). **MVSplat** [23] extends this to sparse multi-view inputs using a cost-volume prior, improving geometric accuracy. **Splatt3R** [24] builds on the DUSt3R framework to predict scene-consistent Gaussians from unposed image pairs.

World-Modelling Implications. Feed-forward 3DGS methods simultaneously address D1 and D2, a significant step. Dynamics (D3) and action conditioning (D4) remain active research targets.

5 Generative Approaches to NVS

5.1 3-D-Aware Generative Adversarial Networks

Before diffusion models, GAN-based methods pioneered the integration of generative priors with 3-D scene representations. **GRAF** [25] and **pi-GAN** [26] condition a radiance-field generator on a latent code, enabling multi-view consistent image synthesis from a single code without real 3-D supervision. **EG3D** [27] introduces the tri-plane representation, encoding a scene on three orthogonal feature planes and decoding with a shallow MLP, dramatically improving quality and training speed.

World-Modelling Implications. These methods demonstrate that strong generative priors can enforce multi-view consistency (D1) while supporting cross-identity generalisation (D2 for objects), but they do not address dynamics (D3) or action conditioning (D4).

5.2 Diffusion-Based NVS

Diffusion models have rapidly become the dominant generative backbone for NVS, owing to their superior sample quality and ability to incorporate conditioning signals.

Zero-1-to-3 [29] fine-tunes a latent diffusion model [28] on multi-view object datasets, conditioning the denoising process on a relative camera pose. At inference, a single RGB image suffices to synthesise novel views (D2), though geometric consistency across multiple synthesised views is limited.

One-2-3-45 [30] addresses this by lifting diffusion multi-view outputs into a consistent 3-D reconstruction, combining the generative prior of Zero-1-to-3 with reconstruction-based consistency.

SyncDreamer [31] enforces 3-D consistency explicitly by jointly denoising multiple views with shared 3-D attention, producing geometrically coherent multi-view outputs from a single image.

Cat3D [32] and **ReconFusion** [33] extend the idea to sparse-view inputs: a multi-view diffusion model generates additional views, which are then fused via NeRF or 3DGS into a complete 3-D scene.

World-Modelling Implications. Diffusion-based methods make substantial progress on D2 (sparse/single-image input) but trade geometric rigour for generative flexibility, leaving a geometry–generation gap.

5.3 Score Distillation Sampling

Score Distillation Sampling (SDS), introduced in **DreamFusion** [34], optimises a 3-D representation (originally NeRF, later 3DGS) so that renders from all viewpoints score highly under a pre-trained 2-D diffusion model. This sidesteps the need for multi-view training data and opens the door to text-driven 3-D generation.

Magic3D [35] and **ProlificDreamer** [36] address the over-smoothing artefacts of vanilla SDS via coarse-to-fine strategies and variational score distillation, respectively. **GaussianDreamer** [37] combines SDS with 3DGS initialisation for faster, higher-quality text-to-3D.

World-Modelling Implications. SDS-based methods achieve open-vocabulary 3-D generation (a form of D2) and begin to encode semantic knowledge, but geometric fidelity remains below reconstruction-based methods.

5.4 Video Diffusion for 4-D NVS

The most direct path to generative world modelling is to unify spatial and temporal generation in a single model.

ViewCrafter [38] conditions a video diffusion model on camera trajectories, generating temporally coherent novel-view sequences from sparse input images. **Cosmos** [39] is a large-scale video generation foundation model trained on internet-scale video, demonstrating emergent 3-D consistency and physical plausibility. **Genie 2** [40] is an interactive world model that generates playable 3-D environments conditioned on actions, directly targeting D4.

World-Modelling Implications. These methods are the closest current approximations to the generalised world model of Eq. (5), partially satisfying D1–D4. Physical plausibility (D5) and long-horizon coherence remain the primary open challenges.

6 Towards World Modelling: 4-D and Beyond

6.1 The 4-D Convergence Thesis

We observe a clear convergence: the NVS community is progressively incorporating the temporal axis, while the world-modelling community is increasingly adopting explicit 3-D representations for spatial grounding. The meeting point is **4-D World Modelling**: a system that generates geometrically consistent, temporally coherent, and physically plausible scene observations under arbitrary viewpoint and action conditions.

Figure 1 illustrates this convergence schematically.

6.2 Embodied and Driving Applications

The practical urgency of world modelling is clearest in embodied AI and autonomous driving.

UniSim [41] models urban driving scenes as neural closed-loop sensor simulators, rendering novel viewpoints for policy training and evaluation. **GAIA-1** [42] learns a generative world model of driving conditioned on ego-actions, video, and text, enabling diverse scenario generation for autonomous vehicle training. Both works underscore that NVS capabilities are the bottleneck: without geometrically faithful viewpoint synthesis, simulated training data fails to transfer to real deployment.

6.3 Interactive World Generation

Genie 2 [40] generates persistent, interactive 3-D environments from a single image prompt, supporting first-person navigation and object interaction. Cosmos [39] scales this to physical world simulation, demonstrating that internet-scale video pre-training can instil emergent 3-D consistency and physics priors. These results suggest that the boundary between NVS and world modelling is already dissolving at the frontier.

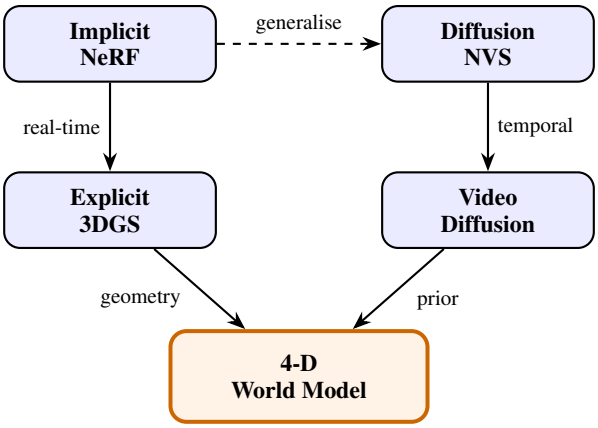

Figure 1: The 4-D Convergence: reconstruction-based NVS (left column) provides geometric rigour; generative NVS (right column) provides visual priors. Both streams are converging toward 4-D World Modelling (bottom).

7 Datasets and Benchmarks

7.1 Static Scene Benchmarks

NeRF-Synthetic [5]: eight synthetic objects with known ground-truth, widely used for controlled NVS evaluation. **LLFF** [6]: forward-facing real scenes; tests interpolation quality. **Tanks and Temples** [44]: large-scale unbounded outdoor and indoor scenes with LiDAR ground truth. **DTU** [45]: multi-view stereo benchmark with object-level scenes and calibrated illumination. **Mip-NeRF 360** [46]: unbounded 360° real scenes with complex backgrounds.

7.2 Dynamic Scene Benchmarks

D-NeRF dataset [10]: six monocular synthetic dynamic scenes. **NHR** [47]: multi-view video of human performance. **CMU Panoptic Studio** [48]: dense multi-view captures of human activities with body pose annotations.

7.3 Large-Scale and Outdoor

Waymo Open Dataset [49] and **nuScenes** [50]: autonomous driving datasets with multi-camera video, LiDAR, and HD maps, used for driving-scene NVS and world modelling.

7.4 Evaluation Metrics and Limitations

Standard NVS metrics—PSNR, SSIM [51], and LPIPS [52]—measure pixel-level or perceptual fidelity to held-out ground-truth views. For world-modelling evaluation, these are insufficient: a system might score highly on PSNR while failing to maintain 3-D consistency, physical plausibility, or action responsiveness. Emerging metrics include **FVD** [53] for temporal coherence and task-specific metrics in robotics (task success rate) and driving (closed-loop collision rate). We identify the absence of a unified world-modelling evaluation suite as a critical gap.

8 Open Problems and Research Roadmap

8.1 The Geometry–Generation Trade-Off

The central tension in the field is between geometric precision and generative flexibility. Reconstruction-based methods (NeRF, 3DGS) deliver accurate geometry but require dense input and encode no generative prior. Diffusion-based methods synthesise plausible views from minimal input but can violate multi-view consistency. **Open question**: Can a single architecture simultaneously achieve reconstruction-quality geometry and diffusion-quality visual priors? Recent work on

3DGS-guided diffusion [32] and diffusion-seeded 3DGS [22] offers partial answers, but a principled solution remains elusive.

8.2 Long-Horizon Temporal Consistency

Existing 4-D methods maintain coherence over short sequences (seconds), but quality degrades sharply over longer horizons due to accumulated error and limited temporal memory. Addressing this requires either explicit 4-D scene memory (*e.g.*, persistent Gaussian maps) or stronger temporal priors (*e.g.*, world model pre-training on long video).

8.3 Physical Grounding

Current NVS methods learn appearance correlations but not physical laws. A world model that understands gravity, collision, or fluid dynamics would generalise far more robustly. Promising directions include integration with physics simulators [43] and energy-based physical priors.

8.4 Scalable Generalisation

Feed-forward 3DGS methods generalise across scenes but are typically trained on specific domains (objects, indoor scenes). Scaling to open-world generalisation (*i.e.*, any scene, any viewpoint) requires internet-scale multi-view data, efficient architectures, and domain-robust training procedures.

8.5 Evaluation Methodology

The field lacks a standard benchmark for world-modelling evaluation. We propose that future benchmarks should assess: (i) geometric consistency under adversarial viewpoint changes; (ii) temporal coherence over long horizons; (iii) action responsiveness in interactive settings; (iv) physical plausibility via physics-engine simulation.

8.6 Research Roadmap

R1. Develop unified architectures that combine explicit 3-D scene representations with diffusion-model priors.

R2. Scale feed-forward 3DGS generalisation to open-world, internet-scale data.

R3. Integrate physical simulators as differentiable priors in 4-D world models.

R4. Design a comprehensive world-modelling evaluation suite spanning geometry, dynamics, action control, and physical plausibility.

R5. Investigate the role of language and semantic grounding in world models as a pathway to action-conditioned generation.

9 Conclusion

This survey has examined Novel View Synthesis through the lens of Generative World Modelling. We traced the evolution of NVS through four eras—classical reconstruction, NeRF, 3DGS, and generative diffusion-based synthesis—and argued that each era represents a step toward the world-modelling desiderata of geometric consistency, generalisation, dynamic modelling, action controllability, and physical plausibility.

Our central thesis—that NVS is the spatial inference engine of world modelling—is substantiated by the convergence of reconstruction-based and generative streams in emerging 4-D methods. Systems such as Genie 2, Cosmos, and ViewCrafter demonstrate that the boundary between NVS and world modelling is already dissolving at the research frontier.

The open problems identified in Section 8—the geometry–generation trade-off, long-horizon consistency, physical grounding, and evaluation methodology—define the research agenda for the next several years. We believe that closing these gaps will yield systems capable of genuine spatial understanding, with transformative implications for robotics, autonomous driving, medical imaging, and interactive content creation.

This is a living draft (v0.1). Content will be expanded in subsequent versions with broader method coverage, richer experimental comparisons, and updated citations.

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